the maximum-mass question, but he now knew that the minimum neutron-core mass was central to Landau's idea.

In his article Landau, also aware of the importance of the minimum neutron-core mass, had used the laws of physics to estimate it. With care Oppenheimer and Serber scrutinized Landau's estimate. Yes, they found, Landau had properly taken account of the attractive forces of gravity inside and near the core. And yes, he had properly taken account of the degeneracy pressure of the core's neutrons (the pressure produced by the neutrons' claustrophobic motions when they get squeezed into tiny cells). But no, he had not taken proper account of the nuclear force that neutrons exert on each other. That force was not yet fully understood. However, enough was understood for Oppenheimer and Serber to conclude that probably, not absolutely definitely, but probably, no neutron core can ever be lighter than $\frac{1}{10}$ of a solar mass. If nature ever succeeded in creating a neutron core lighter than this, its gravity would be too weak to hold it together; its pressure would make it explode.

At first sight this did not rule out the Sun's possessing a neutron core; after all, a $\frac{1}{10}$-solar-mass core, which was allowed by Oppenheimer and Serber's estimates, might be small enough to hide inside the Sun without affecting its surface properties very much (without affecting the things we see). But further calculations, balancing the pull of the core's gravity against the pressure of surrounding gas, showed that the core's effects could not be hidden: Around the core there would be a shell of white-dwarf-type matter weighing nearly a full solar mass, and with only a tiny amount of normal gas outside that shell, the Sun could not look at all like we see it. Thus, the Sun could not possess a neutron core, and the energy to keep the Sun hot must come from somewhere else.

Where else? At the same time as Oppenheimer and Serber in Berkeley were doing these calculations, Hans Bethe at Cornell University in Ithaca, New York, and Charles Critchfield at George Washington University in Washington, D.C., were using the newly developed laws of nuclear physics to demonstrate in detail that nuclear burning (the fusion of atomic nuclei; Box 5.3) can keep the Sun and other stars hot. Eddington had been right and Landau had been wrong—at least for the Sun and most other stars. (As of the early 1990s, it appears that a few giant stars might, in fact, use Landau's mechanism.)